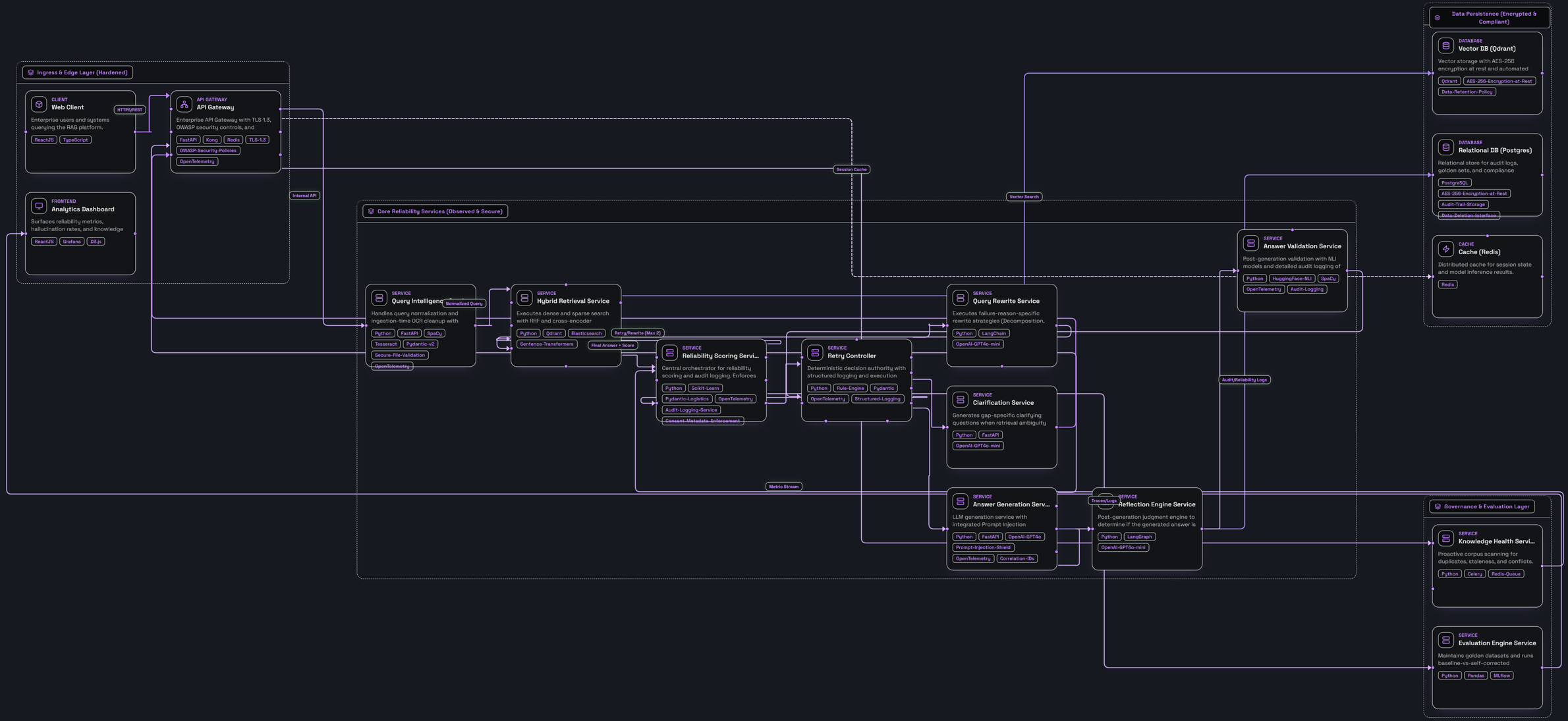
Ingress & Edge Layer (Hardened)
CLIENT
Web Client
Enterprise users and systems querying the RAG platform.
ReactJS
TypeScript
HTTPS/REST
API GATEWAY
API Gateway
Enterprise API Gateway with TLS 1.3, OWASP security controls, and
FastAPI
Kong
Redis
TLS-1.3
OWASP-Security-Policies
OpenTelemetry
FRONTEND
Analytics Dashboard
Surfaces reliability metrics, hallucination rates, and knowledge
ReactJS
Grafana
D3.js
Internal API
Session Cache
Vector Search
Core Reliability Services (Observed & Secure)
SERVICE
Query Intelligenc
Normalized Query
Handles query normalization and ingestion-time OCR cleanup with
Python
FastAPI
SpaCy
Tesseract
Pydantic-v2
Secure-File-Validation
OpenTelemetry
SERVICE
Hybrid Retrieval Service
Executes dense and sparse search with RRF and cross-encoder
Python
Qdrant
Elasticsearch
Sentence-Transformers
Retry/Rewrite (Max 2)
Final Answer + Score
SERVICE
Reliability Scoring Servi...
Central orchestrator for reliability scoring and audit logging. Enforces
Python
Scikit-Learn
Pydantic-Logistics
OpenTelemetry
Audit-Logging-Service
Consent-Metadata-Enforcement
SERVICE
Retry Controller
Deterministic decision authority with structured logging and execution
Python
Rule-Engine
Pydantic
OpenTelemetry
Structured-Logging
SERVICE
Query Rewrite Service
Executes failure-reason-specific rewrite strategies (Decomposition,
Python
LangChain
OpenAI-GPT4o-mini
SERVICE
Clarification Service
Generates gap-specific clarifying questions when retrieval ambiguity
Python
FastAPI
OpenAI-GPT4o-mini
SERVICE
Answer Validation Service
Post-generation validation with NLI models and detailed audit logging of
Python
HuggingFace-NLI
SpaCy
OpenTelemetry
Audit-Logging
Audit/Reliability Logs
Metric Stream
SERVICE
Answer Generation Serv...
LLM generation service with integrated Prompt Injection
Python
FastAPI
OpenAI-GPT4o
Prompt-Injection-Shield
OpenTelemetry
Correlation-IDs
Traces/Logs
SERVICE
Reflection Engine Service
Post-generation judgment engine to determine if the generated answer is
Python
LangGraph
OpenAI-GPT4o-mini
Data Persistence (Encrypted & Compliant)
DATABASE
Vector DB (Qdrant)
Vector storage with AES-256 encryption at rest and automated
Qdrant
AES-256-Encryption-at-Rest
Data-Retention-Policy
DATABASE
Relational DB (Postgres)
Relational store for audit logs, golden sets, and compliance
PostgreSQL
AES-256-Encryption-at-Rest
Audit-Trail-Storage
Data-Deletion-Interface
CACHE
Cache (Redis)
Distributed cache for session state and model inference results.
Redis
Governance & Evaluation Layer
SERVICE
Knowledge Health Servi...
Proactive corpus scanning for duplicates, staleness, and conflicts.
Python
Celery
Redis-Queue
SERVICE
Evaluation Engine Service
Maintains golden datasets and runs baseline-vs-self-corrected
Python
Pandas
MLflow